Consent-Based USB Activity Monitoring System

I am an undergraduate Cyber Security student. I want to develop a **Consent-Based USB Activity Monitoring System** as a cybersecurity practical project using **Python on both Linux and Windows.**

Please act as my **Cyber Security instructor** and guide me from beginning to end.

Project Objective

I want to create a small defensive security application that:

1. Displays a clear **consent notice** before monitoring starts.
2. Starts monitoring only when the user selects **Yes/I Agree.**
3. Detects USB storage device connection and disconnection events.
4. Records only safe activity such as:
    - Date and time
    - USB connected
    - USB disconnected
    - Monitoring started/stopped
    - Copy documents from only directory named “home/documents” in Linux and whatever Documents folder is in Windows to “USB”. Don’t touch any other directory.
5. Saves these events in a log file in usb-monitor/report in user home.
6. Calculates a **SHA-256 hash** of the log file to demonstrate data integrity and save it in usb-monitor/report in user home.
7. Provides a simple report showing the recorded activities and save it in usb-monitor/report in user home.

8. Allows the user to stop monitoring at any time.
9. When program is closed, copy all log files and other data to usb and add data to “usb-monitor” project in relevent directories BUT files copying from documents directory should be direct to USB not in user home and also record its log and file metadata in log.

Strict Privacy and Ethical Requirements

This is an educational cybersecurity project. The application must NOT:

- Steal passwords
- Capture keystrokes
- Collect browser history
- Read browser cookies or saved credentials
- Access webcam or microphone
- Hide itself from the user
- Bypass security
- Automatically execute malicious code from a USB
- Send data secretly to another person
- Monitor a computer without the user's knowledge

Use only a **test/lab computer** and dummy data.

Technology

Use:

- Linux

- Windows
- Python
- Local text/CSV/JSON logging [All should do preferably JSON]
- SHA-256 for integrity verification

How I Want You to Teach Me

Do NOT give me the entire project without explanation.

Build it with me **step by step**, starting from creating the project.

For every step:

1. Tell me exactly what to click or create.
2. Give me the complete code required for that step.
3. Explain the code in simple undergraduate-level language.
4. Tell me where each file/class/function should be placed.
5. Tell me how to run the program.
6. Show me the expected output.
7. Give me a small test to verify that the step works.
8. If an error occurs, explain how to troubleshoot it.

Required Project Structure

Please organize the project professionally, for example:

usb-monitor/

- monitor/
- logs/
- test-data/
- documentation/

For the Python project, use suitable classes such as:

- ConsentManager

- UsbMonitor
- ActivityLogger
- IntegrityManager
- Program/Main

You may change the structure if you have a better design.

Development Stages

Teach me in this order:

Stage 1: Create the Python project.

Stage 2: Create the consent/authorization screen with beautiful screen, no dumb screen like AI slop with nice visuals.

Stage 3: Create the activity logger.

Stage 4: Detect USB connection/disconnection safely.

Stage 5: Record USB events with timestamps.

Stage 6: Create a simple activity report.

Stage 7: Calculate SHA-256 hash of the log.

Stage 8: Demonstrate that modifying the log changes its hash.

Stage 9: Add proper error handling.

Stage 10: Improve the user interface.

Stage 11: Test the complete system.

Stage 12: Explain the cybersecurity concepts demonstrated by the project.

Final Deliverables

At the end, help me prepare:

- Complete source code
- Project folder structure
- Installation/setup instructions
- User manual
- Testing procedure
- Test cases and expected results
- Screenshots I should take
- Short project report
- System architecture diagram
- Flowchart
- Presentation outline
- Viva questions and answers

Important

Keep the project **defensive, transparent, consent-based, and suitable for an undergraduate Cyber Security laboratory.**

If a requested feature would turn the project into spyware, credential theft, covert surveillance, or malware, do not provide that implementation. Instead, explain the security concern and provide a safe laboratory alternative.

Start with **Stage 1: Creating the project**, and wait for me to confirm that it works before moving to Stage 2.